INDIRA GANDHI DELHI TECHNICAL UNIVERSITY FOR WOMEN

CLOUD COMPUTING ASSIGNMENT

Bachelor of Technology
In
Computer Science and Engineering & AI

Submitted By:-

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B.Tech 2nd Year

Submitted To:-

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ASSIGNMENT-01

Question 1

Describe the key computing techniques used in cloud computing such as virtualization, distributed computing, and utility computing.

Answer

Cloud computing relies on several fundamental computing techniques to deliver scalable, reliable, and cost-efficient services over the Internet. The most important among these are Virtualization, Distributed Computing, and Utility Computing.

- **Virtualization**

Virtualization is the technology that allows multiple virtual machines (VMs) to run on a single physical computer. Each virtual machine behaves like an independent system with its own operating system, applications, and resources.

This is achieved using a software layer called a **hypervisor**, which allocates hardware resources such as CPU, memory, storage, and network bandwidth to different virtual machines.

Virtualization improves hardware utilization, reduces costs, and provides flexibility because new virtual machines can be created, modified, or deleted quickly without affecting the physical hardware. It also provides isolation, ensuring that problems in one VM do not impact others.

In cloud computing, virtualization enables cloud providers to host multiple customers on shared infrastructure while maintaining security and independence.

- **Distributed Computing**

Distributed computing refers to a system in which multiple computers located in different places work together through a network to perform a task. Instead of relying on a single powerful machine, the workload is divided into smaller parts and processed simultaneously by multiple machines.

This approach provides high performance, scalability, and reliability. If one computer fails, others can continue processing, ensuring fault tolerance and continuous availability of services.

Cloud data centers consist of thousands of interconnected servers that operate as a distributed system. This allows cloud platforms to handle massive workloads such as big data processing, large-scale web applications, and AI computations efficiently.

- **Utility Computing**

Utility computing is a service model in which computing resources such as processing power, storage, and software are provided on demand and charged based on usage, similar to utilities like electricity or water. Users do not need to purchase or maintain expensive hardware. Instead, they access resources whenever required and pay only for what they use. This makes IT infrastructure more affordable and scalable for businesses of all sizes. Utility computing forms the economic foundation of cloud computing, enabling organizations to scale resources up or down according to their needs while minimizing operational costs.

Question 2

Explain the strengths of cloud computing for organizations and individual users.

Answer

Cloud computing provides many benefits by delivering computing services such as storage, software, and processing power over the Internet.

- **Cost Saving:**
  Users do not need to buy expensive hardware or software. Organizations save on infrastructure and maintenance costs, and individuals can use services at low cost or even free.
- **Scalability:**
  Cloud resources can be increased or decreased according to demand. Businesses can handle heavy workloads easily, and individuals can upgrade storage when needed.
- **Accessibility:**
  Cloud services can be accessed anytime and from anywhere using the Internet. This supports remote work, online learning, and easy access to files on multiple devices.
- **Reliability:**
  Cloud providers use multiple servers, so services remain available even if one server fails. This ensures continuous operation and minimal downtime.
- **Data Backup and Recovery:**
  Data stored in the cloud is automatically backed up. In case of data loss, it can be recovered quickly.
- **Easy Maintenance:**
  Service providers handle updates, security, and maintenance, reducing technical burden on users.
- **Collaboration:**
  Multiple users can work on the same files simultaneously, improving teamwork and productivity.

Question 3

Discuss the limitations and challenges of cloud computing, including security, privacy, and vendor lock-in.

Answer

Although cloud computing provides many advantages, it also has certain limitations and challenges that organizations and individual users must consider.

- **Security Issues:**
  Data stored in the cloud is kept on remote servers, which may be vulnerable to cyberattacks, hacking, malware, or data breaches. Since information is transmitted over the Internet, there is always a risk of unauthorized access. Organizations dealing with sensitive data (like banking or healthcare) must implement strong security measures.
- **Privacy Concerns:**
  In cloud computing, data is stored and managed by third-party service providers. Users may not have full control over how their data is stored or processed. There is also concern about data being stored in different countries, where privacy laws may vary.
- **Vendor Lock-in:**
  Vendor lock-in occurs when a customer becomes dependent on a particular cloud provider and finds it difficult to switch to another provider. This may happen due to differences in technologies, pricing models, or data formats. Migrating data and applications to another provider can be complex and costly.
- **Internet Dependency:**
  Cloud services require a stable and high-speed Internet connection. Without Internet access, users cannot access their data or applications.

- **Downtime and Service Outages:**
  Although cloud providers aim for high availability, service outages can still occur due to technical failures or maintenance issues, affecting business operations.

Question 4

Identify and explain any five application domains of cloud computing (e.g., healthcare, education, e-commerce, banking, big data analytics).

Answer

Cloud computing is widely used in various fields to store data, run applications, and deliver services over the Internet. Five important application domains are:

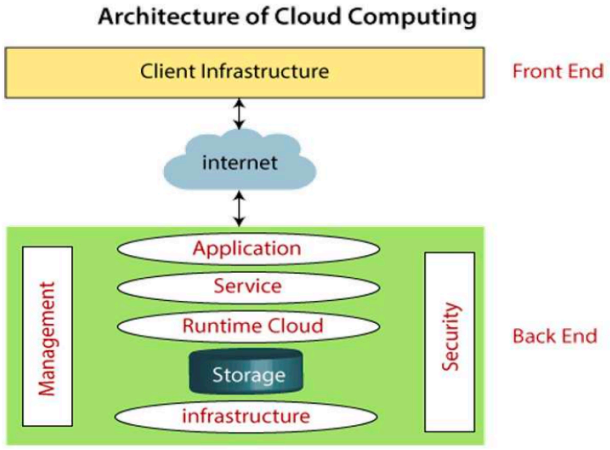
- **Healthcare:**
  Cloud computing is used to store electronic health records (EHRs), medical images, and patient data securely. Doctors can access patient information from anywhere, enabling better diagnosis and treatment. It also supports telemedicine and remote monitoring.
- **Education:**
  Cloud platforms provide online learning systems, virtual classrooms, digital libraries, and storage for study materials. Students and teachers can access resources anytime and collaborate easily through tools like online assignments and video lectures.
- **E-commerce:**
  Online shopping websites use cloud computing to manage product catalogs, customer data, payments, and order processing. The cloud helps handle large traffic during sales and ensures fast and reliable service.
- **Banking and Finance:**
  Banks use cloud services for secure data storage, online banking systems, fraud detection, and transaction processing. It improves service availability and allows customers to access banking services anytime through mobile or web applications.
- **Big Data Analytics:**
  Cloud computing provides powerful processing and storage capabilities needed to analyze massive datasets. Organizations use it to extract insights, predict trends, and support decision-making without investing in expensive infrastructure.

Question 5

Explain the Cloud Computing Architecture with a neat diagram (front-end and back- end components).

Answer

Cloud Computing Architecture refers to the structure of components and technologies that work together to deliver cloud services over the internet. It consists of two main parts: **Front-End** (client side) and **Back-End** (cloud side), connected through a network (Internet).



Cloud computing architecture is divided into the following two parts:
Front End & Back End.

Front End

The front end is used by the client and contains client-side interfaces and applications to access the cloud computing platforms. The front end includes web browsers (including Chrome, Firefox, internet explorer, etc.), thin & fat clients, tablets, and mobile devices.

Thin clients use web browsers facilitating lightweight accessibility, **example:** emails access through web browsers.
Fat clients need strong UX, example: desktop applications which run offline on local servers like MS desktop and MS Outlook.

1. **Client Infrastructure:** provides GUI (Graphical User Interface) to interact with the cloud.

The Internet is a medium through which the front end and back end communicate with each other.

Back End

The back end is used by the cloud service provider. It manages all the resources to provide cloud computing services which includes a huge amount of data storage, security mechanisms, virtual machines, deploying models, servers, traffic control mechanisms, etc.

2. **Application:** software that a client wants to access.
3. **Service Provider:** services such as
   - Software as a Service (SaaS)
   - Platform as a Service (PaaS)
   - Infrastructure as a Service (IaaS)
4. **Runtime Cloud:** provides the execution and runtime environment to the virtual machines.
5. **Storage:** provides a huge amount of storage capacity in the cloud to store and manage data.
6. **Infrastructure:** provides services on the host level, application level, and network level includes components such as servers, storage, network devices, virtualization software, and other storage resources.
7. **Management:** Management is used to manage components such as application, service, runtime cloud, storage, infrastructure, and other security issues in the backend and establish coordination between them.
8. **Security:** is an in-built back end component of cloud computing which implements a security mechanism in the back end.

Question 6

Differentiate between IaaS, PaaS, and SaaS with examples for each service model.

Answer

Parameters	IaaS	PaaS	SaaS
Meaning	Provides virtualized hardware resources over the Internet	Provides a platform to develop, run, and manage applications	Provides ready-to-use software over the Internet
User Control	User controls OS, storage, apps, and data	User controls applications and data only	User only uses the software; no control over platform or infrastructure
Managed By Provider	Networking, storage, servers, virtualization	Infrastructure + OS + runtime	Everything (infrastructure, platform, software)
Purpose	To build and manage complete IT infrastructure	To develop and deploy applications	To use software without installation
Technical Skill Required	High	Moderate	Low
Uses	Hosting websites, virtual servers	App development, testing, deployment	Email, document editing, CRM
Examples	AWS EC2, Microsoft Azure VM, Google Compute Engine	Google App Engine, Heroku, AWS Elastic Beanstalk	Gmail, Google Docs, Microsoft 365, Dropbox

Question 7

What is Infrastructure as a Service (IaaS)? Explain its features and use cases.

Answer

Infrastructure as a a Service (IaaS) is a cloud computing model that provides virtualized computing resources such as servers, storage, and networking over the Internet. Instead of buying physical hardware, users rent infrastructure from a cloud provider on a pay-per-use basis.

In IaaS, the service provider manages the physical hardware and virtualization, while the user controls the operating system, applications, and data.

Features of IaaS

- **On-Demand Self-Service:**
  Users can instantly create, start, stop, or delete virtual machines whenever required without human interaction with the provider. This saves time and increases flexibility.
- **Scalability and Elasticity:**
  Resources can be scaled up during peak demand and scaled down when demand decreases. This ensures smooth performance without overloading systems.

- **Pay-as-You-Go Pricing:**
  Users are billed only for the resources they actually use, such as computing time, storage space, or bandwidth. This reduces capital investment and operational costs.
- **Virtualization Technology:**
  IaaS uses virtualization to divide physical hardware into multiple virtual machines. Each VM operates independently, allowing efficient use of resources.
- **High Availability and Reliability:**
  Cloud providers use multiple servers and data centers with backup systems, ensuring continuous service even if hardware failures occur.
- **Full Control for Users:**
  Users can choose and install their preferred operating system, software, security settings, and configurations according to their needs.

Use Cases of IaaS

- **Website and Application Hosting:**
  Businesses host websites and web applications on virtual servers, avoiding the cost of maintaining physical servers.
- **Development and Testing Environments:**
  Developers can quickly create temporary environments to build, test, and deploy software, then remove them after completion.
- **Data Storage and Backup:**
  Organizations store large volumes of data securely and retrieve it whenever needed. Automatic backup reduces risk of data loss.
- **Disaster Recovery:**
  In case of system failure, cyberattack, or natural disaster, data and applications can be restored quickly, ensuring business continuity.
- **High-Performance Computing and Big Data:**
  IaaS provides powerful processing capabilities required for complex tasks such as scientific simulations, data analytics, and AI workloads.

Question 8

Compare Public Cloud and Private Cloud. Mention their advantages and suitable scenarios.

Answer

Parameters	Public Cloud	Private Cloud
Definition	Cloud infrastructure shared by multiple users and managed by a third-party provider	Cloud infrastructure dedicated to a single organization
Ownership	Owned and managed by cloud service provider	Owned/managed by organization or third party for one organization

Parameters	Public Cloud	Private Cloud
Cost	Lower cost (pay-as-you-go model)	Higher cost due to dedicated resources
Security	Moderate security (shared environment)	Higher security and control
Scalability	Highly scalable	Limited scalability compared to public cloud
Maintenance	Managed by provider	Managed by organization or dedicated provider

Advantages of Public Cloud

- Cost-effective (no need to buy infrastructure)
- Easy scalability for growing workloads
- No maintenance responsibility
- Quick deployment of services

Suitable Scenarios for Public Cloud

- Startups and small businesses with limited budget
- Website hosting and mobile applications
- Development and testing environments
- Applications with variable traffic

Advantages of Private Cloud

- High security and privacy
- Greater control over data and infrastructure
- Customization according to organizational needs
- Suitable for handling sensitive data

Suitable Scenarios for Private Cloud

- Banking and financial institutions
- Healthcare organizations
- Government agencies
- Large enterprises handling confidential data

Question 9

Explain the Hybrid Cloud model. How does it benefit organizations? Give real-world use cases.

Answer

Hybrid Cloud is a cloud computing model that combines both Public Cloud and Private Cloud environments into a single system, allowing data and applications to be shared between them securely. In this model, organizations keep sensitive and critical data in the private cloud for better security and control, while using the public cloud for less sensitive tasks, large-scale processing, or services that require high availability. The two environments are connected through secure networks, enabling smooth data transfer and workload management.

This model provides flexibility, security, and cost efficiency by utilizing the advantages of both cloud types. It also allows organizations to use public cloud resources temporarily during peak demand (cloud bursting) while maintaining core operations in the private cloud.

Benefits to Organizations

- **Improved Security:**
  Confidential data such as financial records, personal information, or trade secrets can be stored in the private cloud where strict security policies are applied. At the same time, public cloud resources can handle general workloads, reducing risk while maintaining performance.
- **Cost Efficiency:**
  Organizations do not need to build large private infrastructures for occasional high workloads. They can use the public cloud only when required, reducing hardware costs, maintenance expenses, and energy consumption.
- **Scalability and Flexibility:**
  Hybrid cloud allows systems to scale easily. During peak periods (e.g., high user traffic), additional resources from the public cloud can be used without upgrading the private cloud infrastructure.
- **Better Control:**
  Organizations retain full control over critical applications and sensitive data in the private cloud while still benefiting from the advanced services and global reach of public cloud providers.
- **Business Continuity:**
  By storing data and applications across both environments, organizations can ensure backup and disaster recovery. If one system fails, operations can continue using the other, minimizing downtime.

Real-World Use Cases

- **Banking and Finance:**
  Banks store sensitive customer and transaction data in private clouds for security and regulatory compliance, while public clouds handle online banking portals, mobile apps, and customer services that require high scalability.
- **Healthcare:**
  Hospitals keep patient medical records in private clouds to maintain privacy, while public clouds support telemedicine platforms, research analysis, and appointment systems accessible to patients.
- **E-commerce Companies:**
  Core systems such as payment databases and inventory management remain in private clouds, while public clouds manage website traffic during major sales events, ensuring smooth user experience.
- **Government Organizations:**
  Sensitive national or citizen data is stored in secure private clouds, whereas public clouds are used to provide online public services like tax filing, information portals, and citizen applications.

Question 10

Explain how virtualization helps cloud providers achieve scalability and cost efficiency. What would happen if virtualization was not used?

Answer

Virtualization is a technology that allows multiple virtual machines (VMs) to run on a single physical server using software called a hypervisor. Each VM behaves like an independent computer with its own operating system and applications.

How Virtualization Provides Scalability

Scalability means the ability to increase or decrease resources according to demand.

- **On-Demand Resource Allocation:** New virtual machines can be created instantly without purchasing new hardware.
- **Elastic Scaling:** CPU, memory, and storage can be increased or decreased dynamically.
- **Load Distribution:** Workloads can be shifted between VMs to handle high traffic.
- **Rapid Deployment:** Services can be launched quickly during peak demand.

Example: During online shopping festivals, cloud providers create additional VMs to handle millions of users and remove them after traffic decreases.

How Virtualization Provides Cost Efficiency

Virtualization significantly reduces operational and infrastructure costs.

- **Efficient Hardware Utilization:** Multiple VMs share a single physical server, reducing idle resources.
- **Reduced Hardware Requirement:** Fewer physical machines are needed.
- **Lower Power and Cooling Costs:** Less equipment means lower electricity consumption.
- **Reduced Maintenance Cost:** Managing fewer physical servers is easier and cheaper.
- **Pay-as-You-Use Model:** Customers pay only for the resources consumed.

Example: One powerful server can replace many underutilized servers.

Without virtualization, cloud computing would become inefficient and expensive:

- Each application would need a separate physical server
- Very high hardware and infrastructure cost
- Poor utilization of resources (many servers remain idle)
- Slow scaling — new hardware must be purchased and installed
- Higher power, cooling, and maintenance expenses
- Reduced flexibility and reliability

Question 11

Explain the concept of Resource Virtualization. How does it help in efficient management of computing resources?

Answer

Resource Virtualization is the technique of creating virtual versions of physical computing resources such as CPU, memory, storage, and network using software. It allows multiple virtual machines (VMs) or users to share the same physical hardware while appearing as separate, independent systems. This is achieved using a software layer called a **hypervisor**, which manages and allocates resources to each virtual machine.

How It Helps in Efficient Management of Computing Resources

1. **Improved Resource Utilization:**
   Physical resources are shared among multiple VMs, ensuring that hardware is not left idle and is used to its maximum capacity.
2. **Dynamic Resource Allocation:**
   Resources such as CPU, RAM, and storage can be allocated or reallocated based on demand, enabling efficient handling of varying workloads.
3. **Scalability:**
   New virtual machines can be created quickly, and resources can be scaled up or down without installing new physical hardware.
4. **Isolation:**
   Each virtual machine operates independently. A failure or heavy load in one VM does not affect others, improving reliability and security.
5. **Cost Efficiency:**
   Fewer physical servers are required, which reduces hardware costs, power consumption, cooling, and maintenance expenses.
6. **Simplified Management:**
   Administrators can monitor and control resources centrally, making management easier and more efficient.

Question 12

What is Server Virtualization? Describe its types and advantages for organizations.

Answer

Server Virtualization is the process of dividing a single physical server into multiple virtual servers using virtualization software called a **hypervisor**. Each virtual server (Virtual Machine — VM) operates independently with its own operating system, applications, and resources, as if it were a separate physical server.

This allows one physical machine to run multiple workloads simultaneously, improving efficiency and flexibility.

Types of Server Virtualization

1. Full Virtualization

In full virtualization, the entire hardware of the physical server is simulated. Each virtual machine runs its own operating system independently without modification.

Example: Running Windows and Linux on the same physical server

Features:

- Complete isolation between VMs
- No need to modify the guest OS
- Easy to manage and widely used

2. Para-Virtualization

In para-virtualization, the guest operating system is modified to interact directly with the hypervisor for better performance.

Features:

- Faster performance than full virtualization

- Requires OS modification
- Less overhead

3. OS-Level Virtualization (Container-Based Virtualization)

In this type, multiple virtual environments run on the same operating system kernel instead of separate OS instances.

Example: Containers like Docker

Features:

- Lightweight and fast
- All environments share the same OS
- Efficient resource usage

Advantages for Organizations

1. **Better Resource Utilization:**
   Multiple virtual servers share one physical server, reducing idle hardware.
2. **Cost Reduction:**
   Fewer physical servers are required, lowering hardware, power, cooling, and maintenance costs.
3. **Scalability and Flexibility:**
   New virtual servers can be created, removed, or modified quickly based on business needs.
4. **Improved Disaster Recovery:**
   Virtual machines can be backed up and restored easily, ensuring business continuity.
5. **Isolation and Security:**
   Problems in one virtual server do not affect others.
6. **Simplified Management:**
   Centralized administration makes monitoring and maintenance easier.

Question 13

Explain Storage Virtualization and its benefits in terms of scalability, performance, and data management.

Answer

Storage Virtualization is the process of combining multiple physical storage devices (such as hard disks and SSDs) into a single virtual storage pool that appears as one logical storage unit to users and applications. This is done using software that hides the physical complexity and manages data efficiently across different storage devices.

In simple terms, many storage devices act like one large storage system.

Benefits of Storage Virtualization

1. Scalability

Storage virtualization allows easy expansion of storage capacity without disrupting existing systems.

- New storage devices can be added to the virtual pool anytime
- No need to stop applications or migrate data manually

- Supports growing data requirements of organizations

Example: If a company needs more space, administrators simply add new disks to the pool.

2. Improved Performance

- Data can be distributed across multiple storage devices, enabling faster read/write operations
- Load balancing prevents any single device from becoming a bottleneck
- Frequently accessed data can be placed on faster storage (e.g., SSDs)

Result: Faster data access and improved system responsiveness.

3. Efficient Data Management

- Centralized control of all storage resources
- Simplifies backup, recovery, and replication processes
- Enables automatic data movement between storage tiers
- Reduces chances of data loss through redundancy mechanisms

Administrators can manage large amounts of data from a single interface.

Question 14

What is Network Virtualization? How does it improve flexibility and security in cloud environments?

Answer

Network Virtualization is the technique of creating multiple virtual networks from a single physical network infrastructure using software. It combines physical network components such as switches, routers, cables, and bandwidth into logical networks that operate independently. Each virtual network has its own IP addresses, routing rules, security policies, and configuration. Thus, one physical network can support many isolated virtual networks simultaneously.

How It Improves Flexibility in Cloud Environments

1. **Rapid Provisioning:**
   Virtual networks can be created, modified, or deleted quickly through software without installing new hardware.
2. **Scalability:**
   Network resources such as bandwidth and connections can be increased or decreased based on demand.
3. **Support for Multi-Tenancy:**
   Multiple organizations (tenants) can share the same cloud infrastructure while having separate private networks.
4. **Easy Configuration and Management:**
   Network settings (IP addressing, routing, policies) can be controlled centrally and changed dynamically.
5. **VM Mobility:**
   Virtual machines can move between physical servers without changing their network configuration, ensuring seamless operation.

How It Improves Security in Cloud Environments

1. **Network Isolation:**
   Each virtual network is logically separated, preventing unauthorized access between users or organizations.
2. **Micro-Segmentation:**
   Security policies can be applied at a very granular level (per VM or application), limiting the spread of attacks.
3. **Virtual Firewalls and Security Controls:**
   Firewalls, intrusion detection, and access controls can be implemented in software without additional physical devices.
4. **Secure Multi-Tenant Environment:**
   Activities of one tenant do not affect others, ensuring confidentiality and data protection.
5. **Reduced Exposure:**
   Sensitive systems can be hidden from public networks and accessed only through controlled channels.

Question 15

Compare VirtualBox and Microsoft Hyper-V in terms of features, platform support, and use cases.

Answer

Parameters	VirtualBox	Microsoft Hyper-V
Features	Type-2 (hosted) hypervisor that runs on top of an existing operating system; supports snapshots, shared folders, USB device access, and multiple guest OSs; easy to install and use.	Type-1 (bare-metal) hypervisor integrated with Windows; provides high performance, virtual switches, dynamic memory allocation, live migration, and advanced networking features.
Platform Support (Host)	Cross-platform — runs on Windows, Linux, macOS, and Solaris.	Limited to Windows Pro, Enterprise, Education editions, and Windows Server.
Guest OS Support	Supports a wide range of operating systems including Windows, Linux, BSD, etc.	Supports Windows and Linux (optimized mainly for Windows environments).
Use Cases	Ideal for personal use, education, software testing, development, and running different OSs on a single PC.	Designed for enterprise environments, data centers, server consolidation, and production workloads.
Performance	Moderate performance since it runs over a host OS.	High performance because it interacts directly with hardware.
Ease of Use	Beginner-friendly with simple interface.	More complex; suited for IT professionals and organizations.